

Question#1 [Coding] [12 marks]

In image processing and data transformation tasks, matrices are often divided into quadrants to manipulate different regions of the data independently. You are given a square matrix of size N x N, where N is always an even number. This allows the matrix to be divided into four equal quadrants.

Your task is to **swap** the elements of the **1st quadrant (Q1)** with the **4th quadrant (Q4)** while keeping the other quadrants unchanged as demonstrated below and **return** the matrix.

Java Method Signature	Python Method Signature																																				
<pre>public static int[][] swapQ1Q4(int[][] matrix){ // To DO }</pre>	<pre>def swapQ1Q4(matrix): //To DO</pre>																																				
Sample Input	Sample Output																																				
<table><tr><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>13</td><td>14</td><td>15</td><td>16</td></tr></table> Matrix Representation: <table><tr><td>1 2 5 6</td><td>3 4 7 8</td></tr><tr><td>9 10 13 14</td><td>11 12 15 16</td></tr></table> where, Q1 = Q4 = 1 2 11 12 5 6 15 16	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	1 2 5 6	3 4 7 8	9 10 13 14	11 12 15 16	<table><tr><td>11</td><td>12</td><td>3</td><td>4</td></tr><tr><td>15</td><td>16</td><td>7</td><td>8</td></tr><tr><td>9</td><td>10</td><td>1</td><td>2</td></tr><tr><td>13</td><td>14</td><td>5</td><td>6</td></tr></table>	11	12	3	4	15	16	7	8	9	10	1	2	13	14	5	6
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Question#2 [Short Questions] [6 marks]

I. What is the fourth last linear index of a matrix of dimensions (3,4,5)? [1 mark] (A) 60 (B) 58 (C) 56 (D) 55
II. Advantage of a singly linked over an array? [1 mark] (A) Random Indexing (B) Fixed Length (C) Dynamic Length (D) Faster Search
III. How do you identify if Node “n” is the second last node in the linked list? [1 mark] Ans:
IV. If you efficiently reverse a linked list(in-place) of 7 nodes, how many nodes do you end up creating? [1 mark] (A) 0 (B) 2 (C) 7 (D) 8
V. What is the linear index of element (3,1,2) in a matrix with dimensions (5,4,3) [1 mark] Ans:
VI. What should the nodeAt() method return for an invalid index? [1 mark] (A) -1 (B) 0 (C) 1 (D) null